
Algorithm 1: Linear Regression Using SSR Loss

Input: X (training data matrix $m \times n$), y (target vector), learning rate α , number of iterations T

Output: w (weights), b (bias)

Initialize w with small random values;

Initialize $b \leftarrow 0$;

$m \leftarrow$ number of samples;

for $t = 1$ **to** T **do**

 // Forward Pass

for $i = 1$ **to** m **do**

$z_i \leftarrow w^\top x_i + b$;

$\hat{y}_i \leftarrow z_i$;

end

 // Compute SSR Loss

$L \leftarrow \sum_{i=1}^m (y_i - \hat{y}_i)^2$;

 // Compute Gradients

$dw \leftarrow \mathbf{0}_n$;

$db \leftarrow 0$;

for $j = 1$ **to** n **do**

$dw_j \leftarrow -2 \sum_{i=1}^m (y_i - \hat{y}_i) x_{ij}$;

end

$db \leftarrow -2 \sum_{i=1}^m (y_i - \hat{y}_i)$;

 // Normalize Gradients

$dw \leftarrow dw/m$;

$db \leftarrow db/m$;

 // Update Parameters

$w \leftarrow w - \alpha \cdot dw$;

$b \leftarrow b - \alpha \cdot db$;

end

return w, b ;

Algorithm 2: Logistic Regression Using Sigmoid Activation

Input: X (training data $m \times n$), y (labels $\in \{0, 1\}$), learning rate α ,
number of iterations T

Output: w (weights), b (bias)

Initialize w with small random values;

Initialize $b \leftarrow 0$;

$m \leftarrow$ number of samples;

Define sigmoid function: $\sigma(z) = \frac{1}{1+e^{-z}}$;

for $t = 1$ **to** T **do**

 // Forward Pass

for $i = 1$ **to** m **do**

$z_i \leftarrow w^\top x_i + b$;

$\hat{y}_i \leftarrow \sigma(z_i)$;

end

 // Binary Cross Entropy Loss

$L \leftarrow \sum_{i=1}^m [-y_i \log(\hat{y}_i) - (1 - y_i) \log(1 - \hat{y}_i)]$;

 // Compute Gradients

$dw \leftarrow \mathbf{0}_n$;

$db \leftarrow 0$;

for $j = 1$ **to** n **do**

$dw_j \leftarrow \sum_{i=1}^m (\hat{y}_i - y_i) x_{ij}$;

end

$db \leftarrow \sum_{i=1}^m (\hat{y}_i - y_i)$;

 // Normalize Gradients

$dw \leftarrow dw/m$;

$db \leftarrow db/m$;

 // Update Parameters

$w \leftarrow w - \alpha \cdot dw$;

$b \leftarrow b - \alpha db$;

end

return w, b ;
